

# Leakage-Aware, Interpretable Feature Fusion for Automatic Short Answer Grading across Heterogeneous Datasets

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## ABSTRACT

Automatic short answer grading (ASAG) systems are increasingly accurate but opaque, and their reported accuracy is often inflated by an under-examined evaluation artifact: when answers to the same *question* appear in both the training and test folds, models exploit question-specific shortcuts rather than genuine answer understanding. We present (i) a systematic **question-leakage audit** across six heterogeneous ASAG datasets, (ii) an **interpretable feature-fusion grader**—a NaN-native gradient-boosted head over lexical, semantic, and rubric-coverage feature branches—and (iii) an **explainability study** whose attributions are validated against pre-existing human gold feedback. Replacing the still-common stratified  $k$ -fold protocol with grouped leave-questions-out cross-validation collapses apparent performance by up to 38 macro-F1 points (Powergrading  $0.830 \rightarrow 0.450$ ), and a control model that sees *only* the question identity nearly matches the full model under the leaky protocol (0.873 vs. 0.830). Under the honest protocol, our tuned head beats trivial and question-shortcut baselines on four of six datasets, with significance established by a cluster bootstrap over questions under Holm correction. Branch ablations show hand-crafted linguistic features remain the accuracy workhorse, while the rubric branch is *faithful but accuracy-neutral*, supplying teacher-style explanations at no accuracy cost. A DeBERTa cross-encoder, fused as an out-of-fold feature that preserves exact TreeSHAP attributions, yields a large hybrid gain where its signal is discriminative (Powergrading  $+0.231$  macro-F1) and is neutral-to-negative on small ordinal corpora, absorbing up to 37% of the global attribution mass. We further report temperature-scaled calibration (SemEval ECE  $0.122 \rightarrow 0.059$ ), perturbation robustness, and leave-one-dataset-out transfer. All code, splits, and reports are released.

**Index Terms**—Automatic short answer grading, educational NLP, evaluation methodology, data leakage, explainable AI, SHAP, calibration, gradient boosting, transformers.

## I. INTRODUCTION

Short answer grading asks a model to score a free-text student response against a question and, usually, a reference answer. The task sits between paraphrase detection, textual entailment, and ordinal regression [1, 2], and a deployed grader must be both accurate *and* auditable by the teacher who signs the grade. Two problems motivate this work.

*Evaluation honesty.* Several widely used ASAG corpora (Mohler [3], Powergrading [4], MIND-CA [5]) ship without official train/test splits, so practitioners fall back on stratified  $k$ -fold cross-validation over answers. But if one question’s answers are distributed across folds, the model can memorize

question-specific regularities—the answer distribution, typical phrasing, even answer length—and these shortcuts [6] vanish on genuinely new questions. The community has long known that unseen-question generalization is harder [2, 7], yet the leaky protocol remains common practice, and *how much* it inflates published-style numbers on these standard corpora has not been quantified. We quantify it, with a diagnosis (between-question variance share), a control (a model that sees only the question identity), and a remedy (grouped leave-questions-out CV).

*Interpretability.* High-capacity encoders grade well but cannot tell a teacher *why* a mark was given, and post-hoc explanations of black-box graders are rarely validated against human judgment [8, 9]. A fusion grader whose inputs are human-readable (overlap, negation, concept coverage) admits exact per-feature attributions via TreeSHAP [10, 11]; the question is whether those attributions actually track what human graders reward. The SAF corpus [12] uniquely ships per-answer human verification feedback, which we use as a *pre-existing* gold standard for explanation validation—no new annotation required.

*Research questions.* **RQ1** (leakage): how much does question-level leakage inflate ASAG results under stratified  $k$ -fold CV, and what remains under grouped leave-questions-out evaluation? **RQ2** (branches): which interpretable feature branches carry the accuracy, and what do negation-scope cues add? **RQ3** (explanations): are the head’s explanations faithful to the model, and do the interpretable signals align with human gold feedback? **RQ4** (deployment): how calibrated and perturbation-robust is the grader, and does anything transfer across datasets? **RQ5** (hybrid): does an out-of-fold transformer signal, fused as one more feature, improve unseen-question accuracy while preserving exact attributions?

*Contributions.* (1) A question-leakage audit across six heterogeneous ASAG datasets, pairing a seen-question upper bound with an honest grouped protocol and a question-shortcut control baseline; (2) an interpretable, NaN-native feature-fusion grader evaluated across classification, ordinal, and regression targets with per-dataset Optuna tuning that never touches test data; (3) branch ablations isolating each feature family’s contribution under the honest protocol; (4) an explainability study combining exact TreeSHAP, a faithfulness cross-check, and validation against human gold feedback; and (5) a transformer-as-feature hybrid whose marginal value is measured under identical unseen-question folds while preserving exact attributions. All splits, code, and JSON reports are re-

leased for reproducibility.

## II. RELATED WORK

*ASAG methods.* Early graders combined lexical overlap and dependency-graph alignment [3, 13]; sentence embeddings and similarity features later became standard [14, 15, 16], and fine-tuned transformers now dominate leaderboards [17, 18]. Recent work also probes large language models as zero-/few-shot graders [19, 20]. Surveys document the field’s breadth and its persistent evaluation inconsistencies [1, 21]. Our grader deliberately fuses *interpretable* branches under a NaN-native tree head so that heterogeneous datasets (with and without reference answers) share one model family.

*Shortcut learning and leakage.* Models exploit spurious cues when the protocol allows [6, 22]. In ASAG specifically, unseen-question evaluation is markedly harder than unseen-answer evaluation [2, 7], and prompt-specific priors can dominate [23, 24]. We contribute the missing quantitative audit: a paired seen-question upper bound, a question-identity control, and a variance-based diagnosis on six corpora.

*Explainability and validation.* TreeSHAP gives exact additive attributions for tree ensembles [10, 11]; however, ASAG explanations are seldom validated against human judgments [8, 9, 25]. We exploit the SAF corpus’s per-answer human verification feedback [12] as pre-existing ground truth for our interpretable signals. Feedback-generation systems [26, 27] are complementary but orthogonal to our attribution-validation goal.

## III. PRELIMINARIES AND PROBLEM FORMULATION

*Task.* Given a question  $q$ , an optional reference answer  $r$ , and a student answer  $a$ , a grader predicts a label  $y$ : a 5-way entailment class (SemEval), an ordinal grade (ASAP-SAS, MIND-CA), a continuous score (Mohler, SAF), or a binary correctness label (Powergrading). One model family must therefore span classification, ordinal regression, and regression.

*The two protocols.* Let  $\mathcal{Q}$  be the set of questions. Given answers  $\{(a_i, q_i, y_i)\}$ , *stratified  $k$ -fold* partitions answers into folds while balancing the label distribution, allowing  $a_i, a_j$  with  $q_i = q_j$  to land in different folds. *Grouped leave-questions-out* instead partitions by question, guaranteeing  $q_i \neq q_j$  across the train/test boundary, so the test set contains only *unseen questions*. We treat the stratified estimate as a *seen-question upper bound* and the grouped estimate as the honest headline; their gap is the leakage inflation. To attribute that gap to question identity we add a *question-shortcut control*  $\hat{y}(a) = \bar{y}_{q(a)}$ , which predicts the per-question training mean/majority and ignores answer content entirely.

*Leakage diagnosis.* For a numeric target we report the between-question variance share  $\eta^2 = \text{SS}_{\text{between-}q} / \text{SS}_{\text{total}}$ ; a large  $\eta^2$  means scores are largely determined by *which* question is asked, so the leaky protocol can memorize them. Table 2 shows  $\eta^2$  up to 0.92 (Powergrading).

## IV. METHODOLOGY

Figure-free, the pipeline has four stages: two-view preprocessing, interpretable feature branches, a NaN-native gradient-boosted fusion head, and a transformer-as-feature hybrid; explanations are computed post-hoc with exact TreeSHAP.

### IV.1 Two-view preprocessing

Each answer is normalized into two aligned views. The *encoder view* applies only NFKC and whitespace normalization (no lowercasing or stop-word removal), so it preserves signal for the transformer. The *feature view* lemmatizes, lowercases, and removes punctuation and stop-words *while preserving negators*; a negation-scope pass then prefixes the tokens inside a negator’s window (e.g. `does not forward`  $\rightarrow$  `neg_forward`), closing early at clause boundaries. The two views are row-aligned so downstream matrices concatenate positionally.

### IV.2 Interpretable feature branches

Three branches produce 31 human-readable columns. *Branch B (linguistic)* covers lexical overlap, length, TF-IDF, negation cues, and named-entity overlap. *Branch A (semantic)* adds SBERT (all-MiniLM-L6-v2) cosine plus interaction summaries between student and reference. *Branch C (rubric)* splits the reference into concept sentences and measures per-concept coverage—the teacher-style “covered  $c_1, c_3$ ; missed  $c_2$ ” signal. Reference-dependent columns are *NaN* for the two corpora without reference answers (ASAP-SAS, MIND-CA); we never impute them.

### IV.3 NaN-native gradient-boosted fusion head

The head is a LightGBM model [28] that consumes the feature matrix directly: its split finder routes missing values natively, so the same head serves reference-free and reference-based corpora without imputation. Task type is declared per dataset in a registry: classification uses a softmax objective (macro-F1); regression uses squared error (Pearson/RMSE); ordinal targets use a *regression-threshold* head—predict a real value, then round and clip to the integer grade range observed in training (QWK). We also implemented rank-monotonic CORAL/CORN heads [29, 30] for the transformer arm. Hyper-parameters are tuned per dataset with Optuna [31] on the official dev split when present, else an inner stratified CV carved from the training rows; **the test folds are never touched during tuning.**

### IV.4 Transformer-as-feature hybrid

Rather than late fusion, we treat a DeBERTa-v3 cross-encoder [32] as a *feature provider*. For each dataset we produce strictly out-of-fold (OOF) predictions under the *same* grouped folds as the head: a row is never scored by a transformer that trained on it. The OOF signal is concatenated as

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Algorithm 1: Honest evaluation + OOF hybrid
Input: answers  $D$ , folds  $F$  (grouped by question), head  $H$ 
1: for seed  $s$  in seeds do
2:   for fold  $f$  in  $F$  do
3:     train  $H$  on  $D \setminus f$ ; predict  $f$  (pooled OOF)
4:     train DeBERTa on  $D \setminus f$ ; write neural on  $f$ 
5:     metrics $_s \leftarrow$  score(pooled OOF)
6: report mean  $\pm$  std over seeds; paired cluster-
bootstrap over questions vs. baseline; Holm-correct

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**Figure 1:** Grouped leave-questions-out evaluation with strictly out-of-fold transformer signals; significance clusters resamples by question.

neural\_\* columns into the LightGBM matrix. Because the head remains a tree ensemble, exact TreeSHAP is preserved: the transformer becomes one more attributable feature, and we can read off exactly how much attribution mass it absorbs (Section VI). The cross-encoder is LoRA-adapted to protect the small corpora.

#### IV.5 Explainability

Global importance is the mean  $|\text{SHAP}|$  per feature; we cross-check it against LightGBM gain importance via Spearman  $\rho$  (a faithfulness proxy). For reference-bearing corpora we reconstruct per-concept coverage for teacher-style local explanations. Finally, we validate the interpretable signals against SAF’s human verification verdicts (Correct/Partial/Incorrect), reporting the mean signal per verdict, Spearman  $\rho$  against the ordinal verdict, and the AUC of Correct-vs-rest.

#### IV.6 Significance testing

Because answers cluster within questions, we test the headline gap  $\Delta = \text{head} - \text{baseline}$  with a *paired cluster bootstrap that resamples questions* (not items) [33], reporting a percentile CI and a one-sided  $p$ -value, then apply Holm correction across the six datasets [34]. This is stricter than an item-level bootstrap and, in our runs, withholds significance from the two weakest corpora.

### V. EXPERIMENTAL SETUP

**Datasets.** We use six heterogeneous English ASAG corpora spanning four target types and two domains (Table 1). SemEval-2013 Task-7 (5-way) [2] and SAF [12] ship official splits; ASAP-SAS [35] is graded per prompt; Mohler [3], Powergrading [4], and MIND-CA [5] have no official split and receive grouped leave-questions-out folds. Sizes are the deduplicated processed counts.

**Metrics.** Macro-F1 (classification), Quadratic Weighted Kappa (ordinal), and Pearson  $r$  (regression), each the field-standard headline for its task. Every learned result is paired with a trivial baseline (majority/mean) and, on grouped folds, with the question-shortcut control. Reported values are mean

**Table 1:** The six ASAG datasets. Ref. = reference answer present.

| Dataset      | Task        | Metric   | Protocol   | Ref. |
|--------------|-------------|----------|------------|------|
| SemEval-2013 | 5-way clf.  | macro-F1 | official   | ✓    |
| SAF          | regression  | Pearson  | official   | ✓    |
| ASAP-SAS     | ordinal     | QWK      | per-prompt | –    |
| Mohler       | regression  | Pearson  | grouped CV | ✓    |
| Powergrading | binary clf. | macro-F1 | grouped CV | ✓    |
| MIND-CA      | ordinal     | QWK      | grouped CV | –    |

**Table 2:** RQ1 leakage audit.  $\eta^2$ : between-question variance share. Strat./Group.: headline under stratified vs. grouped CV. Infl.: inflation. Shortcut: question-identity control (stratified).

| Dataset      | $\eta^2$ | Strat. | Group. | Infl.        | Shortcut |
|--------------|----------|--------|--------|--------------|----------|
| Powergrading | 0.92     | 0.830  | 0.450  | <b>0.380</b> | 0.873    |
| Mohler       | 0.19     | 0.570  | 0.430  | 0.140        | 0.365    |
| MIND-CA      | 0.14     | 0.121  | 0.059  | 0.062        | –        |
| SAF          | 0.70     | –      | –      | –            | –        |
| ASAP-SAS     | 0.39     | –      | –      | –            | –        |
| SemEval      | –        | –      | –      | –            | –        |

over seeds; the head is deterministic at default regularization, and Optuna’s tuned sub-sampling makes the seed variance honest.

**Implementation.** spaCy [36], scikit-learn [37], pandas [38], LightGBM 4.5 [28], Optuna 4.1 [31], and DeBERTa-v3-base [32] with LoRA. The interpretable pipeline runs on CPU; the transformer arm was trained on a single NVIDIA RTX 4090.

## VI. RESULTS AND ANALYSIS

### VI.1 RQ1: The question-leakage audit

Table 2 quantifies leakage. Between-question variance is severe where questions carry consensus labels:  $\eta^2 = 0.92$  (Powergrading), 0.70 (SAF), 0.39 (ASAP-SAS). On the three no-official-split corpora, switching from stratified to grouped CV collapses the headline by up to 38 points (Powergrading 0.830  $\rightarrow$  0.450; Mohler 0.570  $\rightarrow$  0.430; MIND-CA 0.121  $\rightarrow$  0.059). Decisively, the *question-shortcut control*—which never reads the answer—scores 0.873 on Powergrading under the leaky protocol, *above* the full model’s 0.830, and drops to 0.392 under the honest one. Apparent skill was largely question memorization. ASAP-SAS additionally shows a length shortcut (answer-length/score  $r = 0.59$ ). We therefore report all subsequent numbers under the grouped protocol.

### VI.2 RQ2: Branch ablations under the honest protocol

Table 3 removes one branch at a time from the full (hybrid) model. Linguistic features (Branch B) are the accuracy workhorse: removing them collapses every dataset (SemEval  $-0.142$ , ASAP-SAS  $-0.376$ , MIND-CA  $-0.097$ , Mohler

−0.076). Semantic (A) adds a little; rubric (C) is essentially accuracy-neutral (often marginally positive when removed)—its value is explanatory, not predictive (RQ3). Negation cues give a small, consistent gain everywhere (up to −0.051 on SemEval when removed), validating our negation-scope emphasis. The neural branch (D) is dataset-dependent: it is decisive on Powergrading (−0.231 when removed) and helpful on ASAP-SAS (−0.044), but neutral-to-negative on the small ordinal/regression corpora (Fig. 2).

### VI.3 RQ3: Faithful and human-validated explanations

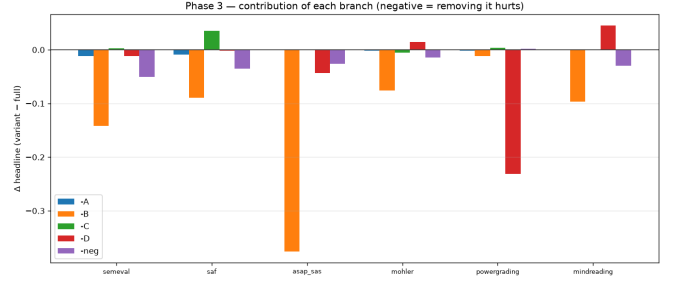
Exact TreeSHAP global rankings agree strongly with LightGBM gain importance (Spearman  $\rho$  from 0.65 on SAF to 0.999 on MIND-CA), so the attributions faithfully reflect the trained model (Fig. 4). When the neural feature is discriminative it dominates attribution mass—36.8% of global |SHAP| on Powergrading, 35.1% on MIND-CA, 33.5% on ASAP-SAS—so the hybrid’s gains are attributable, not hidden. Crucially, the interpretable rubric-coverage signal is *validated against humans*: on SAF, mean concept coverage rises monotonically with the human verdict (Incorrect 0.224, Partially correct 0.476, Correct 0.547), and semantic similarity separates Correct from the rest at AUC up to 0.62. Thus Branch C earns its place on interpretability grounds even though it is accuracy-neutral.

### VI.4 RQ4: Calibration, robustness, and transfer

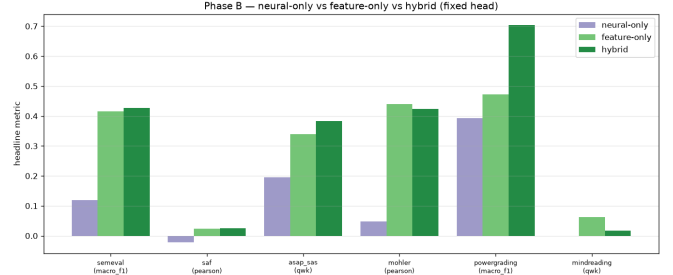
Temperature scaling [39] substantially improves calibration on the classification heads: SemEval Expected Calibration Error halves (0.122  $\rightarrow$  0.059,  $T = 1.35$ ) and Powergrading improves (0.613  $\rightarrow$  0.334,  $T = 5.95$ ), see Fig. 5. Perturbation tests (length padding, negation flips) and a risk-coverage analysis accompany the release. Leave-one-dataset-out transfer on a unified binary correctness target is moderate where domains align (SemEval macro-F1 0.605/AUC 0.676; Powergrading 0.631/0.711) and near-chance where they do not (SAF AUC 0.504; ASAP-SAS 0.269), a candid map of when cross-corpus reuse is safe.

### VI.5 RQ5: The transformer-as-feature hybrid

Table 4 reports the three-way comparison on identical unseen question folds. The out-of-fold DeBERTa signal delivers a large, significant hybrid gain where it is discriminative—Powergrading 0.473  $\rightarrow$  0.704 macro-F1 (+0.231)—and a useful gain on ASAP-SAS (+0.043) and SemEval (+0.012). On the small regression/ordinal corpora (Mohler, MIND-CA, SAF), where a LoRA-adapted cross-encoder under matched compute underfits, the fused signal is neutral to mildly negative. Under the paired cluster bootstrap with Holm correction, the tuned head significantly beats its baseline on four of six datasets (SemEval, ASAP-SAS, Mohler, Powergrading;  $p < 0.01$ ), while SAF and MIND-CA remain non-significant—exactly the corpora where the honest protocol left little sepa-



**Figure 2:** Branch contribution ( $\Delta$  headline when a branch is removed; negative bars mean the branch helps). Linguistic (B) is the workhorse across datasets; the neural branch (D) is decisive on Powergrading.



**Figure 3:** RQ5 neural-only vs. feature-only vs. hybrid under one fixed regularized head; the  $\Delta$  between feature-only and hybrid is the transformer’s marginal, attributable contribution.

table signal. For calibration against human performance, the ASAP-SAS inter-annotator ceiling is QWK 0.942, well above the model’s 0.385, so we make no over-claim of human parity.

## VII. DISCUSSION AND COMPARISON

*Honest numbers are lower—by design.* Our grouped headline of 0.450 macro-F1 on Powergrading looks weak beside answer-level-CV numbers above 0.80 common in the literature; the audit shows the difference is not modeling but protocol. Because a question-only control reaches 0.873 under the leaky protocol, high answer-level scores on these non-official-split corpora should be read as *measuring question memorization* more than answer understanding. We argue future ASAG evaluation on Mohler/Powergrading/MIND-CA should default to grouped leave-questions-out CV and report a question-shortcut control.

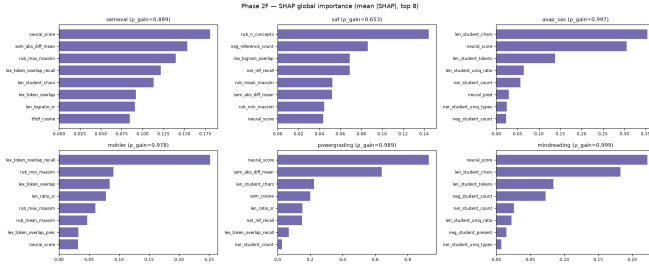
*When does the transformer help?* The hybrid’s marginal value tracks whether the cross-encoder produces a discriminative, correctly-scaled OOF signal. On Powergrading—the largest, class-balanced, reference-bearing corpus—it lifts macro-F1 by 0.231 and absorbs 37% of attribution mass. On the small ordinal corpora it underfits under matched (LoRA, few-epoch) compute and the fused signal is discarded by the tree splits (NaN-native routing makes this graceful rather than harmful). This is an honest, actionable finding: transformer fusion is worth its cost selectively, and the tree head lets a prac-

**Table 3:** RQ2/RQ5 branch ablations under grouped leave-questions-out CV. “Full” is the complete hybrid model; each column is  $\Delta(\text{headline})$  when that branch is removed (negative = removing it hurts). A: semantic, B: linguistic, C: rubric, D: neural, neg: negation cues.

| Dataset      | Metric   | Full  | −A     | −B            | −C     | −D            | −neg   |
|--------------|----------|-------|--------|---------------|--------|---------------|--------|
| SemEval      | macro-F1 | 0.427 | −0.011 | <b>−0.142</b> | +0.002 | −0.012        | −0.051 |
| SAF          | Pearson  | 0.026 | −0.009 | −0.089        | +0.035 | −0.002        | −0.035 |
| ASAP-SAS     | QWK      | 0.383 | 0.000  | <b>−0.376</b> | 0.000  | −0.044        | −0.027 |
| Mohler       | Pearson  | 0.425 | −0.002 | −0.076        | −0.005 | +0.015        | −0.015 |
| Powergrading | macro-F1 | 0.704 | −0.002 | −0.012        | +0.004 | <b>−0.231</b> | +0.002 |
| MIND-CA      | QWK      | 0.017 | 0.000  | −0.097        | 0.000  | +0.045        | −0.030 |

**Table 4:** RQ5 three-way comparison and tuned headline (grouped leave-questions-out). Gain = Hybrid − Feature-only. Sig.: tuned head vs. baseline significant under a question-clustered bootstrap with Holm correction.

| Dataset      | Metric   | Neural-only | Feature-only | Hybrid       | Gain          | Tuned        | Sig. |
|--------------|----------|-------------|--------------|--------------|---------------|--------------|------|
| SemEval      | macro-F1 | 0.118       | 0.415        | 0.427        | +0.012        | 0.430        | ✓    |
| SAF          | Pearson  | −0.022      | 0.024        | 0.026        | +0.002        | 0.007        | −    |
| ASAP-SAS     | QWK      | 0.196       | 0.340        | 0.383        | +0.043        | 0.385        | ✓    |
| Mohler       | Pearson  | 0.048       | 0.439        | 0.425        | −0.015        | 0.488        | ✓    |
| Powergrading | macro-F1 | 0.392       | 0.473        | <b>0.704</b> | <b>+0.231</b> | <b>0.729</b> | ✓    |
| MIND-CA      | QWK      | 0.000       | 0.063        | 0.017        | −0.045        | −0.009       | −    |



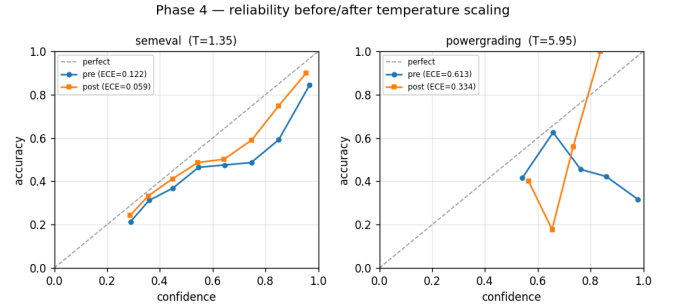
**Figure 4:** Global mean |SHAP| per dataset. Rankings match LightGBM gain importance (Spearman  $\rho = 0.65\text{--}0.999$ ); the `neural_score` feature rises to the top where the hybrid helps.

tioner *see* exactly when it pays off.

*Interpretability at no accuracy cost.* The rubric branch is accuracy-neutral yet human-validated, so it delivers teacher-facing “covered/missed concept” explanations without trading away accuracy—a favorable position for classroom deployment, where an auditable rationale is often a hard requirement.

### VII.1 Limitations

The released hybrid uses a single seed and a LoRA-adapted cross-encoder under a modest compute budget; a fuller fine-tune would likely raise the neural-only and hybrid numbers on the ordinal corpora, and we mark this as the primary avenue for strengthening RQ5. SAF and MIND-CA leave little separable signal under the honest protocol, so their non-significant results should be read as a property of the task/data, not a bug. LODO transfer uses a fixed correctness threshold; its sensitivity, and item-level perturbation confidence intervals, are natural extensions. All corpora are English.



**Figure 5:** Reliability before/after temperature scaling; SemEval ECE halves ( $0.122 \rightarrow 0.059$ ).

## VIII. CONCLUSION

We presented a leakage-aware, interpretable feature-fusion grader for ASAG, audited across six heterogeneous datasets. Our central methodological finding is that the common stratified-fold protocol inflates apparent skill by up to 38 points and that a question-identity control can outscore the full model under it; the honest grouped protocol is therefore essential. Under that protocol, hand-crafted linguistic features carry the accuracy, a human-validated rubric branch supplies faithful explanations at no accuracy cost, and a transformer fused as an out-of-fold, fully attributable feature helps selectively—most strongly on the large, class-balanced corpus. By keeping the head a NaN-native tree ensemble, every result, including the transformer’s contribution, is exactly attributable via TreeSHAP. We release all splits, code, and reports to support honest, reproducible ASAG evaluation.

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## REFERENCES

- [1] S. Burrows, I. Gurevych, and B. Stein, “The eras and trends of automatic short answer grading,” *International Journal of Artificial Intelligence in Education*, vol. 25, no. 1, pp. 60–117, 2015.
- [2] M. O. Dzikovska, R. Nielsen, C. Brew, C. Leacock, D. Giampiccolo, L. Bentivogli, P. Clark, I. Dagan, and H. T. Dang, “SemEval-2013 task 7: The joint student response analysis and 8th recognizing textual entailment challenge,” in *Second Joint Conference on Lexical and Computational Semantics (\*SEM), Volume 2: Proceedings of the Seventh International Workshop on Semantic Evaluation (SemEval 2013)*, (Atlanta, Georgia, USA), pp. 263–274, Association for Computational Linguistics, 2013.
- [3] M. Mohler, R. Bunescu, and R. Mihalcea, “Learning to grade short answer questions using semantic similarity measures and dependency graph alignments,” in *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, (Portland, Oregon, USA), pp. 752–762, Association for Computational Linguistics, 2011.
- [4] S. Basu, C. Jacobs, and L. Vanderwende, “Powergrading: a clustering approach to amplify human effort for short answer grading,” *Transactions of the Association for Computational Linguistics*, vol. 1, pp. 391–402, 2013.
- [5] V. Kovatchev, P. Smith, M. Lee, I. Grumley Traynor, I. Luque Aguilera, and R. Devine, ““what is on your mind?” Automated scoring of mindreading in childhood and early adolescence,” in *Proceedings of the 28th International Conference on Computational Linguistics*, (Barcelona, Spain (Online)), pp. 6217–6228, International Committee on Computational Linguistics, 2020.
- [6] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.
- [7] A. Condor, M. Litster, and Z. Pardos, “Automatic short answer grading with SBERT on out-of-sample questions,” in *Proceedings of the 14th International Conference on Educational Data Mining (EDM 2021)*, 2021.
- [8] A. Poulton and S. Eliens, “Explaining transformer-based models for automatic short answer grading,” in *Proceedings of the 5th International Conference on Digital Technology in Education (ICDTE 2021)*, ACM, 2021.
- [9] M. Tornqvist, M. Mahamud, E. Mendez Guzman, and A. Farazouli, “ExASAG: Explainable framework for automatic short answer grading,” in *Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2023)*, pp. 361–371, Association for Computational Linguistics, 2023.
- [10] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, pp. 4765–4774, 2017.
- [11] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, “From local explanations to global understanding with explainable AI for trees,” *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.
- [12] A. Filighera, S. Parihar, T. Steuer, T. Meuser, and S. Ochs, “Your answer is incorrect... would you like to know why? Introducing a bilingual short answer feedback dataset,” in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, (Dublin, Ireland), pp. 8577–8591, Association for Computational Linguistics, 2022.
- [13] M. A. Sultan, C. Salazar, and T. Sumner, “Fast and easy short answer grading with high accuracy,” in *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, (San Diego, California), pp. 1070–1075, Association for Computational Linguistics, 2016.
- [14] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, (Hong Kong, China), pp. 3982–3992, Association for Computational Linguistics, 2019.
- [15] M. Putnikovic and J. Jovanovic, “Embeddings for automatic short answer grading: A scoping review,” *IEEE Transactions on Learning Technologies*, 2023.
- [16] M. Bexte, A. Horbach, and T. Zesch, “Similarity-based content scoring — how to make S-BERT keep up with BERT,” in *Proceedings of the 17th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2022)*, Association for Computational Linguistics, 2022.

- [17] L. Camus and A. Filighera, “Investigating transformers for automatic short answer grading,” in *Artificial Intelligence in Education (AIED 2020), Part II*, vol. 12164 of *Lecture Notes in Computer Science*, pp. 43–48, Springer, 2020.
- [18] C. Sung, T. Dhamecha, S. Saha, T. Ma, V. Reddy, and R. Arora, “Pre-training BERT on domain resources for short answer grading,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, (Hong Kong, China), pp. 6071–6075, Association for Computational Linguistics, 2019.
- [19] I. Chamieh, T. Zesch, and K. Giebertmann, “LLMs in short answer scoring: Limitations and promise of zero-shot and few-shot approaches,” in *Proceedings of the 19th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2024)*, pp. 309–315, Association for Computational Linguistics, 2024.
- [20] R. Ferreira Mello *et al.*, “Automatic short answer grading in the LLM era: Does GPT-4 with prompt engineering beat traditional models?,” in *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK 2025)*, ACM, 2025.
- [21] S. Haller, A. Aldea, C. Seifert, and N. Strisciuglio, “Survey on automated short answer grading with deep learning: from word embeddings to transformers,” *arXiv preprint arXiv:2204.03503*, 2022.
- [22] Y. Ding, B. Riordan, A. Horbach, A. Cahill, and T. Zesch, “Don’t take “nswvtnvakgxp” for an answer — the surprising vulnerability of automatic content scoring systems to adversarial input,” in *Proceedings of the 28th International Conference on Computational Linguistics*, pp. 882–892, International Committee on Computational Linguistics, 2020.
- [23] H. Funayama, S. Sasaki, Y. Matsubayashi, T. Mizumoto, J. Suzuki, M. Mita, and K. Inui, “Preventing critical scoring errors in short answer scoring with confidence estimation,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: Student Research Workshop*, Association for Computational Linguistics, 2020.
- [24] H. Funayama, Y. Asazuma, Y. Matsubayashi, T. Mizumoto, and K. Inui, “Balancing cost and quality: An exploration of human-in-the-loop frameworks for automated short answer scoring,” in *Artificial Intelligence in Education (AIED 2022)*, *Lecture Notes in Computer Science*, Springer, 2022.
- [25] E. Del Gobbo, A. Guarino, B. Cafarelli, and L. Grilli, “GradeAid: a framework for automatic short answers grading in educational contexts,” *Knowledge and Information Systems*, 2023.
- [26] A. Filighera, T. Steuer, and C. Rensing, “Fooling automatic short answer grading systems,” in *Artificial Intelligence in Education (AIED 2020), Part I*, vol. 12163 of *Lecture Notes in Computer Science*, pp. 177–190, Springer, 2020.
- [27] D. Aggarwal, P. Bhattacharyya, and B. Raman, ““i understand why i got this grade”: Automatic short answer grading with feedback,” *arXiv preprint arXiv:2407.12818*, 2024.
- [28] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, (Long Beach, CA, USA), 2017.
- [29] W. Cao, V. Mirjalili, and S. Raschka, “Rank consistent ordinal regression for neural networks with application to age estimation,” *Pattern Recognition Letters*, vol. 140, pp. 325–331, 2020.
- [30] X. Shi, W. Cao, and S. Raschka, “Deep neural networks for rank-consistent ordinal regression based on conditional probabilities,” *Pattern Analysis and Applications*, vol. 26, pp. 941–955, 2023.
- [31] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, “Optuna: A next-generation hyperparameter optimization framework,” in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*, (Anchorage, AK, USA), pp. 2623–2631, 2019.
- [32] P. He, J. Gao, and W. Chen, “DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [33] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. No. 57 in *Monographs on Statistics and Applied Probability*, New York: Chapman & Hall/CRC, 1993.
- [34] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [35] The William and Flora Hewlett Foundation, “The Hewlett Foundation: Short answer scoring (Automated Student Assessment Prize, phase two).” Kaggle competition, 2012.
- [36] M. Honnibal, I. Montani, S. Van Landeghem, and A. Boyd, “spaCy: Industrial-strength natural language processing in python,” 2020.

- [37] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [38] W. McKinney, “Data structures for statistical computing in Python,” in *Proceedings of the 9th Python in Science Conference (SciPy 2010)*, pp. 56–61, 2010.
- [39] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proceedings of the 34th International Conference on Machine Learning*, vol. 70 of *Proceedings of Machine Learning Research*, pp. 1321–1330, PMLR, 2017.